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Assessing the impact of stand structure on forest net primary productivity: A multiple machine learning-SHAP models and DSEM integrated approach

Tao Li, Yi Wu, Fang Ren, Lei Tian, Mingyang Li^{*}

Co-Innovation Center for Sustainable Forestry in Southern China, Nanjing Forestry University, Nanjing, China

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ABSTRACT

Forest Net Primary Productivity (NPP) reflects the productive capacity of a vegetation community under natural conditions and serves as a key indicator of sustainable forest management. The stand structure determines the competitive potential and spatial niche of trees, significantly influencing forest productivity. Understanding how various stand structure factors influence forest NPP is essential for optimizing forest management measures. To enhance model interpretability, this study employs SHapley Additive exPlanations (SHAP), which leverage Shapley values from game theory to assess feature contributions. SHAP not only identifies key variables but also quantifies their direction and magnitude of impact. This study combines multiple machine learning feature selection methods with SHAP to determine the importance and influence of different stand structure factors on forest NPP. However, the influence of stand structure on forest NPP changes with time. Analyzing this relationship at a single time point may be insufficient. To address this, the study applies a Dynamic Structural Equation Model (DSEM) to capture temporal variations, providing a more comprehensive understanding of how stand structure influences forest productivity over time. The results show that: (1) both stand spatial structure and non-spatial structure have a significant impact on forest NPP; (2) the impact of forest species on forest NPP is the strongest, with a SHAP value mean > 3.2 , followed by stand density, with a SHAP value mean > 1.6 ; (3) the impact of the stand non-spatial structure (path coefficients = $0.71/-0.28$) on forest NPP is stronger than that of the stand spatial structure (path coefficients = $0.51/-0.37$); (4) the influence of stand structure on forest NPP exhibits a lag effect and varies with time. This study highlights the crucial role of both spatial and non-spatial stand structure factors in regulating forest NPP. The findings suggest that forest management measures should prioritize both stand density and species composition to enhance forest productivity. Additionally, recognizing the dynamic nature of stand structure's influence on NPP with time will help in developing more effective, adaptive management practices for sustainable forest management.

1. Introduction

As a major component of terrestrial ecosystems (Le Quéré et al., 2009), forest ecosystem accounts for approximately two-thirds of annual carbon sequestration in terrestrial ecosystems and plays an irreplaceable role in regulating the global carbon cycle, mitigating the increase of greenhouse gases, and maintaining global climate stability (Alexandrov et al., 1999; Zhang et al., 2023; Ma et al., 2021). Forest Net Primary Productivity (NPP) refers to the amount of organic matter accumulated per unit area and unit time, representing the organic carbon fixed through photosynthesis minus the portion consumed by plant respiration (Chapin et al., 2006; Zhou et al., 2022). Forest NPP serves as a foundational metric for studying the flow of matter and energy within

ecosystems, reflecting the productive capacity of vegetation communities under natural environmental conditions and directly influencing the carbon sequestration of the forest ecosystem (Clark et al., 2001; Xu et al., 2020). Forest NPP is not only one of the key indicators of sustainable forest ecosystem management but also a major factor in assessing ecosystem carbon sinks and regulating ecological processes (McMurtrie and Dewar, 2013; Gim et al., 2017; Guillemot et al., 2017).

Stand structure determines the competitive potential and spatial niche of trees, and therefore largely influences the stability, development trajectory, and productivity potential of forests (Prior and Bowman, 2014; Weng et al., 2019; Liu et al., 2020). Stand structure influences and limits forest productivity and carbon sequestration functions (Berzaghi et al., 2023). Stand structure includes spatial and

^{*} Corresponding author.

E-mail address: lmy196727@njfu.edu.cn (M. Li).

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non-spatial components, where spatial structure mainly refers to the distribution, mixing, and competition levels of trees within the stand; non-spatial structure encompasses factors such as stand age, canopy density, canopy height, and average diameter at breast height (DBH). In previous studies, forest stock volume and biomass were first calculated based on the average DBH and canopy height of the stand, and forest NPP was then estimated using stand age (Fang et al., 2002; Wang et al., 2018), or alternatively, forest NPP was directly estimated using stand age (Li et al., 2024). It can thus be seen that stand structure has a direct impact on forest NPP. This demonstrates that stand structure has a direct impact on forest NPP. Despite this understanding, most existing studies have primarily focused on the influence of individual factors on forest NPP (Wang et al., 2011; He et al., 2012; Xia et al., 2019; Bongers et al., 2021; Ma et al., 2021). Few research, however, has investigated how each specific factor within the stand structure affects changes in forest NPP, the direction of this influence (positive or negative), and the relative importance of these factors.

Feature variable selection allows for the identification of all independent variables that can influence the dependent variable (Theng and Bhojar, 2024). However, aside from linear models, many existing machine learning-based nonlinear feature selection methods are considered to be black-box models. Linear models have the advantages of being easy to use and highly interpretable (McGibney, 2023). Linear feature selection relies on the assumption of a linear relationship between the dependent and independent variables. If the relationship is nonlinear, the selection results may be biased (Lederer, 2022). Linear models are highly sensitive to outliers (Muthukrishnan and Rohini, 2016), and the presence of outliers in the data may affect the final selection results. Linear models are flexible and offer simplicity and interpretability. However, their application is limited by sensitivity to outliers, the assumption of independence, restricted capacity to model complex relationships, multicollinearity issues, and difficulty in capturing nonlinear relationships (Joshi, 2023). Nonlinear machine learning models have started to be widely used because they overcome the limitations of linear models. However, compared to linear models, nonlinear machine learning models generally have lower interpretability (Saranya and Subhashini, 2023). To enhance the interpretability of machine learning models, we combined traditional black-box nonlinear machine learning methods with SHapley Additive exPlanations (SHAP). Based on the concept of Shapley values from game theory, SHAP provides a way to fairly allocate cooperative gains among participants. In machine learning, SHAP calculates each feature's marginal contribution across different combinations, ultimately determining each feature's contribution to model predictions. This approach quantifies the impact of each feature on model outputs, improving interpretability and offering deeper insights into the model's decision-making process (Lundberg and Lee, 2017; Parisineni and Pal, 2024). In this study, SHAP is used to explain the relationships between each independent variable and the dependent variable, as well as to assess the relative importance of the independent variables.

The Structural Equation Model (SEM), which is based on the covariance matrix of variables, describes causal relationships between them by establishing a system of linear equations (Catherine et al., 2012; Hair et al., 2021). The model can not only handle relationships involving multiple causes and multiple outcomes but also accommodate unobservable latent variables. The influence of many factors on forest NPP is a long-term process, with causal relationships often exhibiting a lag effect (VanderWeele et al., 2020; Lüdtke and Robitzsch, 2022). Therefore, using data from only a single point in time to investigate the relationship between stand structure and forest NPP may result in errors. This study introduces a Dynamic Structural Equation Model (DSEM) to investigate the impact of stand structure on forest NPP over a period of ten years. DSEM combines the advantages of various modelling approaches, including multilevel models, time series models, structural equation models, and time-varying effect models (Asparouhov et al., 2018; Hamaker et al., 2023). DSEM enables a more convenient and

comprehensive analysis of the relationships between latent and observed variables, latent variables and latent variables, as well as observed variables and observed variables within the model (Hamaker et al., 2018; Zhou et al., 2021). DSEM can be used to model and analyse not only two-variable models but also those with three or more variables.

Current research mainly focuses on external factors influencing changes in forest NPP, such as climate and soil. However, stand structure, which includes factors like tree density, height, and diameter distribution, may directly affect forest NPP and could serve as a more effective predictor (Dănescu et al., 2016; Ali and Mattsson, 2017; Yuan et al., 2018). Despite its importance, the role of stand structure in driving forest NPP, particularly its dynamic and lagged effects, remains under-explored in the research field of forest management. To address this research gap, this study introduces a novel approach that integrates machine learning with SHAP to quantify the impact of stand structure factors on forest NPP, while employing a Dynamic Structural Equation Model (DSEM) to further investigate the temporal dependencies (lag effects) of these factors. This methodological combination allows for a comprehensive analysis of both immediate and delayed impacts of stand structure on NPP, offering new insights into the temporal dynamics of forest ecosystems. The specific objectives of this study are: (1) To assess how different stand structure factors influence forest NPP. (2) To quantify the relative importance of these stand structure factors in determining forest NPP. (3) To explore the lagged effects of stand structure factors on forest NPP. Based on these objectives, we propose the following research hypotheses: (1) Stand structure factors significantly and directly influence forest NPP. (2) Non-spatial stand structure factors exert a stronger influence on forest NPP than spatial factors. (3) The influence of stand structure factors on forest NPP exhibits a lagged effect. By integrating machine learning with DSEM, this study aims to test these hypotheses and provide a comprehensive understanding of how stand structure dynamically influences forest NPP. The findings are expected to offer valuable insights for forest management and regional climate change mitigation strategies.

2. Data and methods

2.1. Research area and data source

Subtropical forests play a crucial role in global ecosystems, contributing significantly to terrestrial carbon cycling and carbon storage (Zareen et al., 2023). China's subtropical forests are globally unique, characterized by diverse forest types, high species richness, and high productivity (Ouyang et al., 2020; Wu et al., 2020). Located in south-eastern Eurasia, these forests are influenced by the monsoon climate, experiencing frequent typhoons and heavy rainfall in summer, and serve key ecological functions such as water conservation and soil erosion control (Ma et al., 2016). Guangdong Province, a key subtropical forest region in China, is vital for timber production and carbon sequestration. As part of the southern collective forest region, it features a high proportion of collectively owned forest land, fragmented forest parcels, short harvest cycles, and complex stand structures (Xu, 1990; Ma et al., 2020). This study focuses on Guangdong Province (20°09'~25°31'N, 109°45'~117°20'E) to explore the impact of stand structure on forest NPP in subtropical regions (Fig. 1).

This study used data from the National Forest Continuous Inventory (NFCI) of Guangdong Province for the years 2002, 2007, and 2012. The NFCI is conducted at the provincial level using a systematic sampling design, with fixed sample plots established at 6 km × 8 km intervals across the province. A total of 3,685 fixed plots were set up, each covering an area of 0.067 ha. The plots were surveyed every five years to ensure consistent and comparable data over time (Lei et al., 2009; Zeng et al., 2015; Wang et al., 2020). Within each plot, multiple stand-level attributes were recorded, including the average age, average diameter at breast height (DBH), and average height of the stand. These attributes

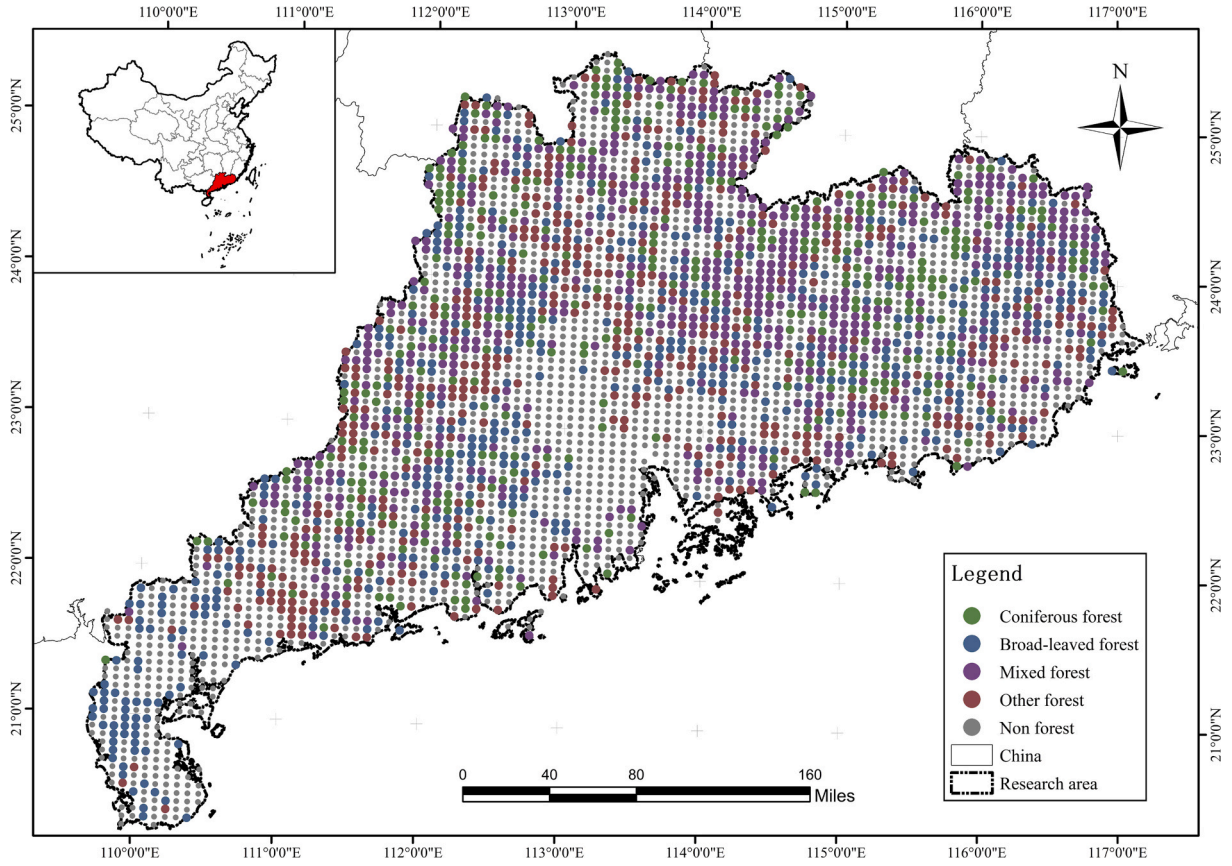


Fig. 1. Location of the study area and the fixed sample plots of NFCI by forest types.

represent the non-spatial structure information and were directly obtained from the plot-level survey results. Additionally, detailed information on each tree, such as its relative position, azimuth, and DBH, was collected, forming the individual tree-level data. These tree-level data were used to derive spatial structure information, such as tree distribution patterns and spatial relationships among trees. Non-spatial stand attributes (e.g., average age, DBH, and height) were imported into ArcGIS for spatial visualization and analysis, including vectorization to create spatial representations of the surveyed areas (Fig. 1). Spatial stand structure factors were calculated using R software based on individual tree measurements. Specifically, we calculated spatial indices, including Mingling Index, Competition Index, Size Ratio, Uniform Angle Index, and Stand Density, to analyze the spatial structure of the forest stands (sample plots).

2.2. Methods

2.2.1. Calculation of spatial structure factors and forest NPP

This study selected 10 non-spatial and spatial stand structure factors to represent stand structure. The required non-spatial stand structure factors (forest species, average stand height, average canopy density, average stand age, and average DBH) were directly obtained from the NFCI attribute table. The spatial stand structure primarily includes three aspects: species spatial segregation, tree size differentiation, and spatial distribution patterns (Hui et al., 2019). Species spatial segregation was described using the mingling index (Hu and Hui, 2015), tree size differentiation was characterized by the Hegyi competition index (Hegyi, 1974) and the size ratio (Hui et al., 1999a), while spatial distribution patterns were quantified using the uniform angle index (Hui et al., 1999b) and stand density (Vanderschaaf, 2013).

(1) Mingling Index

The mingling index is defined as the proportion of neighboring trees around a reference tree that belong to different species from the reference tree (Hui and Hu, 2001), and is expressed by the formula:

$$M_i = \frac{1}{n} \sum_{j=1}^n v_{ij} \quad (1)$$

where, M_i represents the mingling index of tree i ; n is the number of nearest neighboring trees. When the reference tree i and the j -th nearest neighbor belong to different species, v_{ij} is assigned a value of 1; if they belong to the same species, v_{ij} is assigned a value of 0. Hui et al. (2001) suggested that $n = 4$ is sufficient for analyzing the spatial structure of mixed forests. Therefore, $n = 4$ is used in this study.

The mingling index calculated using formula (1) is referred to as the tree-level mingling index (M_i). The stand-level mingling index (M) is the average of the tree-level mingling indices, and its calculation formula is as follows:

$$M = \frac{1}{N} \sum_{i=1}^N M_i \quad (2)$$

where, M represents the stand-level mingling index; N is the number of trees within the stand; and M_i is the tree-level mingling index of the i -th tree.

(2) Competition Index

The individual tree competition index is a quantitative measure of the intensity of competition among trees, reflecting the relationship between the growth of individual trees and their available space (Yang et al., 2019). As trees grow, the available space continuously changes, leading to competition for or sharing of limited resources, which may

result in mutual promotion, or growth suppression, or even tree mortality. This phenomenon is known as tree competition. This study selected the Hegyi competition index proposed by Hegyi (1974) to measure the tree competition, and the formula is:

$$CI_i = \sum_{j=1}^n \frac{D_j}{D_i \times Dist_{ij}} \quad (3)$$

where, CI_i represents the simple competition index of the reference tree i ; D_i is the diameter of the reference tree i ; D_j is the diameter of the j -th competitor tree around the reference tree (where $j = 1, 2, \dots, n$); and $Dist_{ij}$ is the distance between the reference tree i and the competitor tree j .

(3) Size ratio

The size ratio (neighborhood comparison) is defined as the proportion of neighboring trees that are larger than the reference tree, relative to all the nearest neighbors considered. It can be applied to metrics such as DBH, tree height, and crown width (Hui et al., 1999a). This study selects DBH as the indicator. The mean size ratio for each trees indicates the dominance level of that trees with respect to the measured stand indicators and can be calculated using the following formula:

$$U_i = \frac{1}{N} \sum_{i=1}^N K_{ij} \quad (4)$$

$$U = \frac{1}{N} \sum_{i=1}^N U_i \quad (5)$$

where, U_i represents the size ratio of the tree; N is the number of nearest neighboring trees. When the neighboring tree j is smaller than the reference tree i , K_{ij} is assigned a value of 0; otherwise, it is assigned a value of 1. Hui et al. (2003) suggested that $N = 4$ is sufficient for analyzing the spatial structure of mixed forests. A smaller U value indicates a higher priority of the species in a given comparison indicator (such as diameter at breast height, tree height, or crown width).

(4) Angular scale

The uniform angle index proposed by Hui (1999b) is calculated based on n nearest neighboring trees. It can effectively assess variability among groups, even for relatively small clusters. Starting from the reference tree, there are two possible angles between any two nearest neighboring trees, let the smaller angle be α . The uniform angle index UAI_i is defined as the proportion of these angles α that are smaller than the standard angle α_0 among the nearest n neighboring trees considered (Hui et al., 2016), as given by the following formula:

$$UAI_i = \frac{1}{n} \sum_{i=1}^N z_{ij} \quad (6)$$

$$UAI = \frac{1}{N} \sum_{i=1}^N UAI_i = \frac{1}{4N} \sum_{i=1}^N \sum_{j=1}^4 z_{ij} \quad (7)$$

where, N represents the number of reference trees within the stand; i denotes any reference tree; j refers to the four nearest neighboring trees of reference tree; and UAI_i is the uniform angle index, which describes the uniformity of the neighboring trees around reference tree i . As the overall uniform angle index for the stand, UAI is the average of all UAI_i values across the stand and provides a single measure of tree distribution uniformity. When the j -th angle α is smaller than the standard angle α_0 , $z_{ij} = 1$; otherwise, $z_{ij} = 0$. Typically, $n = 4$ and $\alpha_0 = 72^\circ$ (Hui et al., 1999).

(5) Stand density

In this study, stand density is represented by basal area, which is calculated as the total cross-sectional area of trees per unit area (Avery and Burkhart, 2002). Unlike the Stand Density Index (Vanderschaaf, 2013), which may not be applicable to all forest stands, basal area provides a more universally applicable measure of stand density across different forest types. It directly reflects both tree size and density, making it a more flexible and informative metric for assessing stand structure and resource competition.

$$BA_i = \frac{\pi D_i^2}{4} \quad (8)$$

$$SD = \frac{\sum BA_i}{A} \quad (9)$$

where, BA_i represents the basal area of an individual tree; D_i is the diameter at breast height; A is the stand area; SD represents the total cross-sectional area of trees at breast height per unit area.

(6) Calculation of forest NPP

Forest NPP consists of community growth (the sum of the annual net growth of stems, branches, and roots in the upper tree layer and the annual net growth in the lower shrub and herbaceous layer) and annual litterfall (Yu et al., 2014). Forest NPP is calculated based on the relationship between biomass and volume for different forest types, as well as the functional relationships between biomass, community growth, and annual litterfall. The specific formula is:

$$B = \frac{V}{a + b^*V} \quad (10)$$

$$Y = \frac{B}{c^*A + d^*B} \quad (11)$$

$$L = \frac{B}{e + f^*B} \quad (12)$$

$$NPP = Y + L \quad (13)$$

where, B represents biomass, measured in t/ha; V represents stand volume, also measured in m³/ha; Y denotes community growth, in units of t · ha⁻¹ · a⁻¹; L stands for annual litterfall, in t · ha⁻¹ · a⁻¹; A is the stand average age; NPP represents the net primary productivity of the forest, measured in t · ha⁻¹ · a⁻¹; a, b, c, d, e, f are coefficients specific to different forest types (Wang et al., 2010). See the Appendix for specific calculation formulas by forest types.

2.2.2. Feature variable selection

(1) Boruta-SHAP

Boruta is a global feature selection method based on Random Forest (RF) importance evaluation. It assesses feature significance by comparing real variables with randomly generated “shadow variables” (Kursa et al., 2010a; Manikandan et al., 2024). The method works by perturbing original features (Iranzad and Liu, 2024) and introducing shadow features—shuffled copies of real features—before training a Random Forest model (Kursa et al., 2010b). Boruta effectively handles non-linear relationships and complex interactions while minimizing the risk of omitting important variables.

SHapley Additive exPlanations (SHAP) is a method based on Shapley values from game theory that explains machine learning model outputs (Benedek et al., 2022; Chen et al., 2023; Rodrigues et al., 2023). It quantifies each feature’s marginal contribution to the model’s prediction, offering both global and local interpretability (Bokati et al., 2024). SHAP provides an additive explanation, where the predicted output equals the sum of individual feature contributions (Lundberg and Lee,

2017). This property makes SHAP well-suited for complex, non-linear models, enabling it to capture feature interactions (Emanuele et al., 2024). Additionally, differences in SHAP values indicate the relative importance of features.

SHAP is suitable for complex black-box models. Therefore, this study combines Boruta and SHAP to not only filter variables but also quantify each feature's contribution to the model's prediction, enhancing interpretability. Boruta-SHAP ensures that all selected features contribute to the prediction and make the model's decision process more interpretable.

(2) Random Forest Feature Importance - SHAP (RF-SHAP)

The feature selection embedded in RF is based on the frequency of splits and the importance scores of features within the decision trees (Breiman, 2001), typically calculated by measures such as Gini importance or Mean Decrease in Accuracy. Therefore, RF-SHAP inherits the robustness of RF in handling noise and non-linear relationships in data, making it suitable for large datasets without requiring assumptions of linear relationships between variables. Additionally, it provides an intuitive explanation of the importance of feature variables to model predictions, enhancing interpretability.

(3) eXtreme Gradient Boosting – SHAP (XGBoost-SHAP)

XGBoost is an ensemble learning algorithm based on gradient-boosted decision trees, determining feature importance by accumulating each feature's split count and split gain across all trees (Zieba et al., 2016; Song et al., 2023). XGBoost-SHAP combines XGBoost's strengths—such as its ability to handle large datasets, manage missing data, and deliver fast performance—with SHAP's ability to intuitively explain the importance of feature variables in model predictions.

(4) Recursive Feature Elimination – SHAP (RFE-SHAP)

RFE begins by training a model using all available features and calculating each feature's weight or coefficient. It then ranks features based on their weights or coefficients, typically considering those with the smallest weights or coefficients as least important. The least important feature(s) are removed, and the model is retrained with the remaining features. This process of ranking and removing features is repeated iteratively until the desired number of features is reached or no further removal is possible (Kuhn and Johnson, 2013). RFE's advantages include effectively reducing overfitting, enhancing model generalization, and improving efficiency and accuracy by selecting only the most important features (Ding et al., 2022).

RFE alone does not explain the selected features or indicate the degree of their importance. Therefore, after identifying the most important features, SHAP values are used to explain each feature's contribution to the final model output, quantifying each feature's marginal contribution to predictions. RFE-SHAP helps reduce redundant features, simplify the model, decrease training time, lower the risk of overfitting, and interpret the contribution of RFE-selected features, thereby helping to understand the model's predictions.

In this study, we selected four ensemble feature selection models and combined them with SHAP to interpret the selected features. This approach enables comparison of feature selection results across multiple models and assesses the importance of each feature within a single model's results. By identifying the key factors that most significantly influence forest NPP, we gain a deeper understanding of which spatial structure factors (e.g., species mixing degree, competition) and non-spatial factors (e.g., tree age, DBH) have substantial impacts, thereby providing a basis for forest management and policy-making. Additionally, the use of multiple feature selection methods allows for a comparison of how each method defines important features, providing a more comprehensive understanding of the relationship between forest

NPP and feature variables. While different methods may yield slightly varied lists of important features, comparing these results helps us identify the most critical and stable features, thus enhancing model generalizability. To compare the influence of various features on NPP, we employed different feature selection models to enhance result interpretability. Specifically, we used RF in the Boruta-SHAP and RF-SHAP models, and XGBoost in the XGBoost-SHAP and RFE-SHAP models. To mitigate the effects of randomness, each model was trained multiple times. When the training reached over 30 runs, the mean feature importance scores had stabilized. To further improve the stability and reliability of the results, reduce the impact of random fluctuations, and ensure greater accuracy in the feature importance scores, this study conducted 50 training runs, averaging the feature importance scores to enhance the robustness of the feature selection results.

2.2.3. Driving factor analysis

Dynamic Structural Equation Modelling (DSEM) combines the characteristics of Structural Equation Modelling (SEM) and time series analysis to study complex relationships between variables, particularly in multilevel or dynamic environments (Asparouhov et al., 2018). Dynamic Structural Equation Modelling (DSEM) fully leverages the advantages of Structural Equation Modelling (SEM), allowing the study of complex causal relationships between multiple latent variables (Grace and Bollen, 2008; Bollen et al., 2022). DSEM extends SEM by constructing complex path models that explain causal relationships within temporal dynamics through observed and latent variable relationships (Thorson et al., 2024). This capability for time series analysis, which captures evolving dynamic relationships over time, represents DSEM's primary improvement over traditional SEM (Hamaker et al., 2018). For instance, the effect of one variable on another may vary over time, and DSEM can model and explain this temporal change. Additionally, DSEM allows for the modelling of random effects to capture individual differences and also accommodates fixed effects to analyze factors exerting consistent influence across all individuals over time (McNeish et al., 2024). Furthermore, DSEM can handle temporal dynamics not only at the individual level but also at the group level or higher hierarchical levels.

The DSEM decomposes the observed variable into three components, expressed as follows:

$$Y_{it} = Y_{1,it} + Y_{2,i} + Y_{3,t} \quad (14)$$

$Y_{1,it}$ is the within-individual time-varying component, capturing time-specific effects for each individual and reflecting within-individual fluctuations over time. It includes lagged effects from previous time points and captures dynamic interactions among multiple variables. $Y_{2,i}$ is the between-individual component, representing stable individual-level differences over time. It captures consistent variability among individuals, often modeled with random effects to reflect these time-invariant characteristics. $Y_{3,t}$ is the between-time component, addressing variations across time that impact all individuals similarly. It reflects environmental or contextual factors affecting the entire samples at each time point.

$$Y_{1,it} = \nu_1 + \sum_{l=0}^L \lambda_{1,l} \eta_{1,i,t-l} + \sum_{l=0}^L R_l Y_{1,i,t-l} + \sum_{l=0}^L K_{1,l} X_{1,i,t-l} + \varepsilon_{1,it} \quad (15)$$

$$\eta_{1,it} = \alpha_1 + \sum_{l=0}^L B_{1,l} \eta_{1,i,t-l} + \sum_{l=0}^L Q_l Y_{1,i,t-l} + \sum_{l=0}^L \Gamma_{1,l} X_{1,i,t-l} + \xi_{1,it} \quad (16)$$

$$Y_{2,i} = \nu_2 + \lambda_2 \eta_{2,i} + K_2 X_{2,i} + \varepsilon_{2,i} \quad (17)$$

$$\eta_{2,i} = \alpha_2 + B_2 \eta_{2,i} + K_2 X_{2,i} + \xi_{2,i} \quad (18)$$

$$Y_{3,t} = \nu_3 + \lambda_3 \eta_{3,i} + K_3 X_{3,i} + \varepsilon_{3,i} \quad (19)$$

$$\eta_{3,i} = \alpha_2 + B_3\eta_{3,i} + K_3X_{3,i} + \xi_{3,i} \quad (20)$$

where, λ is the factor loading matrix, and R , K , B , Q and Γ are regression coefficient matrices. $\varepsilon_{1,it}$, $\xi_{1,it}$, $\varepsilon_{2,i}$, $\xi_{2,i}$, $\varepsilon_{3,i}$ and $\xi_{3,i}$ represent normally distributed errors with a mean value of zero. $\eta_{i,t}$ represents the latent variable of the i -th individual at time point t . ν and α are constants. $X_{1,i,t-1}$ represents the covariate of the i -th individual at the previous time point $t-1$.

Compared to traditional SEM, DSEM better captures temporal correlations and dynamic changes between variables. It also handles multiple time series data to explain how prior states influence subsequent states while accounting for dynamic changes at both individual and group levels. In this study, we used forest structure factors and forest NPP data from 2002, 2007, and 2012 to explore the causal relationships between structure factors and forest NPP using DSEM model implemented in Mplus 8.

This study employs Multiple Machine Learning-SHAP to quantify the contribution of individual stand structure factors to NPP predictions while using DSEM to model their dynamic interactions over time. SHAP ensures transparency in variable importance, whereas DSEM provides insights into the causal and temporal dependencies among factors. The combination of these methods enables a more interpretable and ecologically meaningful assessment of forest productivity drivers.

3. Results and analysis

3.1. Variable selection results and analysis

Based on NFCI data, all arbor forest sample plots in the study area were selected, and five non-spatial structure factors were directly obtained from the attribute table. Additionally, five spatial structure factors and forest NPP were calculated, with results shown in Fig. 2. Considering the differing magnitudes of the factors, all 10 stand structure factors were normalized. According to Fig. 2, both the stand structure and forest NPP in the sample plots changed over time. Among the spatial structure indices, Mingling index (M) showed the greatest change, with a significant increase in its mean value. Similarly, the mean value of Angular scale (UAI) exhibited an upward trend, while Competition index (CI), stand Size ratio (U), and Stand Density (SD) displayed only minor fluctuations. Regarding the non-spatial structure, the distribution of average stand age (AGE) shifted from initially concentrated to increasingly dispersed, which maybe due to human-induced harvest activities. Trees that were not harvested continued to grow and reached higher AGE values, whereas newly trees planted after harvesting showed lower AGE values, leading to a more varied age distribution. Meanwhile, the mean value of Forest Species (FS) increased significantly, shifting from a dispersed to a more concentrated distribution, which suggests that the study area gradually transitioned from pure coniferous and broad-leaved forests to mixed coniferous-broadleaf forests. Additionally, the mean values of diameter at breast height (DBH), average canopy density (FD), and average stand height (FH) showed slight increases. Overall, the mean forest NPP exhibited a clear upward trend over time.

The results of feature selection are influenced not only by the selection methods but also by the modelling approaches, as each is based on different assumptions that build their feature-target relationships.

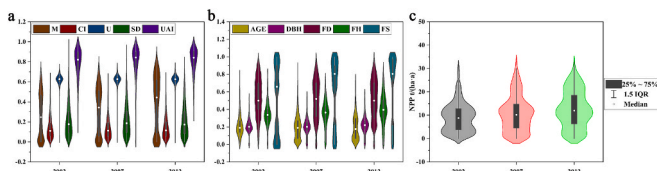


Fig. 2. The value distribution of stand structure factors and forest NPP over time.

Both Random Forest and XGBoost are tree-based models that assess feature importance through contributions to node splits, favoring features that significantly reduce impurity. However, due to inherent randomness—such as Random Forest's tree generation through random sampling and XGBoost's randomization in split selection—different training processes may yield varying feature selections.

As shown in Table 1, all ten stand structure factors were selected regardless of the method used, indicating that both spatial and non-spatial structure factors influence NPP. Further analysis of the feature ranking and importance values of different methods reveals slight variations. In the Boruta-SHAP model, the ranking is FS (3.209) > SD (1.669) > FH (1.052) > AGE (0.989) > FD (0.852) > DBH (0.671) > CI (0.282) > M (0.198) > UAI (0.165) > U (0.081), while in RF-SHAP model, the variable ranking is the same, though importance values differ. The XGBoost-SHAP model shows a change in ranking: FS (3.416) > SD (2.210) > AGE (1.652) > DBH (1.192) > FD (0.920) > FH (0.742) > CI (0.294) > M (0.264) > UAI (0.227) > U (0.199). Similarly, RFE-SHAP model produces the same ranking as XGBoost-SHAP, but with different importance values. This consistency in rankings between Boruta-SHAP and RF-SHAP (both using Random Forest) and between XGBoost-SHAP and RFE-SHAP (both using XGBoost) suggests that feature selection results are influenced by both the selection and modelling methods.

Based on the results of four selection methods, FS consistently ranks as the most important factor, with importance values significantly higher than those of other factors. Photosynthesis, the process by which plants convert solar energy into organic matter, is the core source of NPP. Variations in leaf morphology, chlorophyll content, and stomatal regulation across tree species lead to differences in photosynthetic efficiency, meaning that changes in FS directly impact forest NPP. Additionally, tree species differ in water and nutrient use efficiency. For example, certain species have deeper root systems, enabling them to absorb groundwater and soil nutrients more effectively, thereby supporting higher biomass production levels and directly influencing NPP.

CI, UAI, M, and U, as spatial structure factors, consistently show lower importance compared to other factors, whereas SD, also a spatial structure factor, maintains a higher importance value. This suggests that appropriate stand density contributes to more efficient utilization of light, water, and nutrients, thereby significantly improving forest productivity. While factors such as species spatial segregation (M) and tree size differentiation (CI and U) have relatively weaker impacts on NPP.

AGE, DBH, FH, and FD consistently rank high across different selection and modelling methods. Although these four factors retain the same relative ranking across feature selection methods, the order of

Table 1

The mean absolute SHAP importance of stand structure factors selected using different selection methods.

Feature	Boruta-SHAP	RF-SHAP	Feature	XGBoost-SHAP	RFE-SHAP
	RF (Modelling methods)			XGBoost (Modelling methods)	
FS	3.209	3.211	FS	3.416	3.414
SD	1.669	1.699	SD	2.210	2.276
FH	1.052	1.048	AGE	1.652	1.636
AGE	0.989	0.987	DBH	1.192	1.189
FD	0.852	0.836	FD	0.920	0.938
DBH	0.671	0.644	FH	0.742	0.686
CI	0.282	0.279	CI	0.294	0.293
M	0.198	0.201	M	0.264	0.257
UAI	0.165	0.170	UAI	0.227	0.208
U	0.081	0.080	U	0.199	0.194

Notes: FS represents forest species, SD represents stand density, FH represents average stand height, AGE represents average stand age, FD represents forest average canopy density, DBH represents diameter at breast height, CI represents competition index, M represents mingling index, UAI represents angular scale, U represents stand size ratio.

importance varies across different models, indicating that feature importance is related to the model, indicating that feature importance is model-dependent. Feature selection offers an initial filtering results, but the ultimate importance of variables depends on the selected modelling method. Therefore, selecting an appropriate modelling approach is very crucial; relying solely on one specific method may lead to biased results, highlighting the need for a multi-model comparison.

Additionally, comparing the importance of spatial and non-spatial factors reveals that non-spatial structure factors exert a stronger influence on forest NPP than spatial factors. Non-spatial structural factors (such as tree age, average DBH, and tree height) directly reflect the growth status and health conditions of individual trees in the forest, and these factors have a direct impact on photosynthetic efficiency and carbon fixation capacity. Additionally, non-spatial structural factors involve physiological characteristics and resource acquisition capabilities of trees, such as root structure and the absorption of water and nutrients, significantly influencing resource use efficiency of individual trees. Therefore, compared to spatial structural factors, non-spatial structural factors have a more important impact on NPP.

Apart from showcasing the importance of each feature, SHAP plots in Fig. 3 also illustrate how each sample's feature factors affect forest NPP. Taking AGE as an example, over 70 % of samples show a positive impact on NPP, while a smaller proportion show a negative impact. This reflects the age-NPP relationship of young-aged forest: young trees grow rapidly, allocating resources primarily to new leaves and roots, which improves photosynthetic efficiency and thus has a positive effect on NPP. In the mid-aged forest sample plots, trees allocate more energy to lignified tissues, such as trunks, rather than fast-growing leaves or roots. Biomass accumulation continues to increase, but NPP growth may slow. In the mature or over mature-aged forest sample plots, tree growth slows significantly, with more energy allocated to maintaining existing biomass rather than new growth. Although total biomass (and thus carbon storage) is high, the number of newly formed tissue is very limited, and photosynthetic efficiency declines, resulting in a negative impact of increased age on NPP.

3.2. Drivers results and analysis

This study used forest structure factors and forest NPP data from three time points (2002, 2007, and 2012) to establish a Dynamic Structural Equation Model (DSEM). The model explores the effects of latent variables, including spatial and non-spatial forest structure factors, on forest NPP, as well as the lagged impacts of latent variables on

observed variables. In the 'Original Model' (Fig. 4), we considered not only the influence of latent variables on the dependent variable but also the relationships among independent variables. However, an excessive number of independent variables and paths increases model complexity, which can complicate result interpretation and lead to issues like reduced degrees of freedom, unstable parameter estimates, and overfitting. Therefore, iterative optimizations to the original model are necessary.

Collinearity tests indicate that there is no multicollinearity among the 10 independent variables; however, high correlations among certain variables can lead to model overfitting (Fig. 5). To prevent information loss from variable deletion, we initially attempted dimension reduction using Principal Component Analysis (PCA) on the highly correlated variables. However, the resulting model still showed signs of overfitting and yielded a non-positive definite covariance matrix, which obscured causal relationships and complicated interpretation. Therefore, in the 'Optimized Model' (Fig. 4), we directly removed U (highly correlated with CI), UAI (highly correlated with SD), and FH and AGE (both highly correlated with DBH). Among the 'Optimized Model' fit indices, $\chi^2/df = 1.659$, CFI = 0.932, RMSEA = 0.023, NFI = 0.942, and CFI = 0.973, indicate satisfactory model performance.

Path coefficients indicate causal relationships and the strength of influence among variables. The 'Optimized Model' shows that non-spatial structure is most strongly influenced by FS (0.68), followed by FD (0.48) and DBH (0.23), all with positive impacts. Spatial structure is primarily influenced by SD (0.53), with additional positive effects from CI (0.23) and M (0.12). FS also positively influences M (path coefficient 0.41), while DBH positively affect SD (path coefficient 0.17). This suggests an interdependent relationship between non-spatial and spatial structure, as changes in non-spatial structure drive variations in spatial structure. Further analysis reveals a correlation of 0.54 between the two latent variables. Non-spatial structure has a path coefficient of 0.71 with forest NPP, while spatial structure has a coefficient of 0.51, indicating both types of factors positively affect NPP, though non-spatial structure has a stronger impact.

DSEM enables time series analysis, capturing the dynamic relationships, such as those between spatial structure, non-spatial structure, and forest NPP, which influence the forest NPP in next observation period (i.e., the subsequent NPP measure). Forest NPP in the current period also impacts NPP in the following periods. However, path coefficients show that spatial structure has a stronger effect on the NPP in the latter period than non-spatial structure, but with a negative influence (as shown in Fig. 4). Since this study uses data with a five-year interval, growth may intensify competition for resources, meaning that beneficial stand structures in the current period may lead to heightened competition or other adverse effects in the next period, suppressing NPP growth. As shown in Fig. 6, there is a lag effect of stand structure factors on forest NPP. This suggests that current changes in stand structure not only affect NPP in the present period but will also continue to effect forest productivity over time. Based on these results, it is evident that the influence of both spatial and non-spatial stand structures on forest NPP is dynamic, with the effect changing in intensity and even the direction over time. Furthermore, the magnitude of the lag effect itself also fluctuates with time.

4. Discussion

Current studies have primarily focused on the impact of external factors, such as climate and soil, on forest NPP (LeBauer and Treseder, 2008; Zhang and Zhou, 2011; Liang et al., 2015; Wang et al., 2020; Li et al., 2022). Undoubtedly, forest NPP is influenced by climatic factors like temperature and rainfall, as well as soil properties such as texture and nutrient content. However, Arshad et al. (2018) suggests that climate and soil do not directly impact forest biomass but exert influence indirectly through stand structure. From an ecological perspective, climatic conditions such as temperature, precipitation, and sunlight

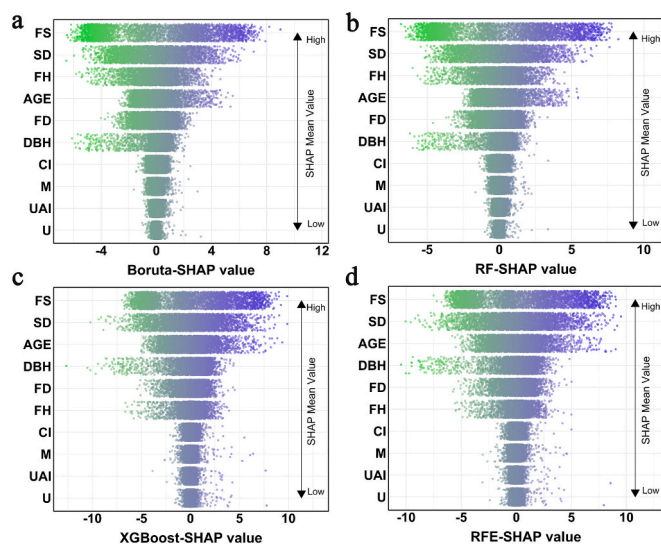


Fig. 3. Importance of stand structure factors obtained from different feature selection methods.

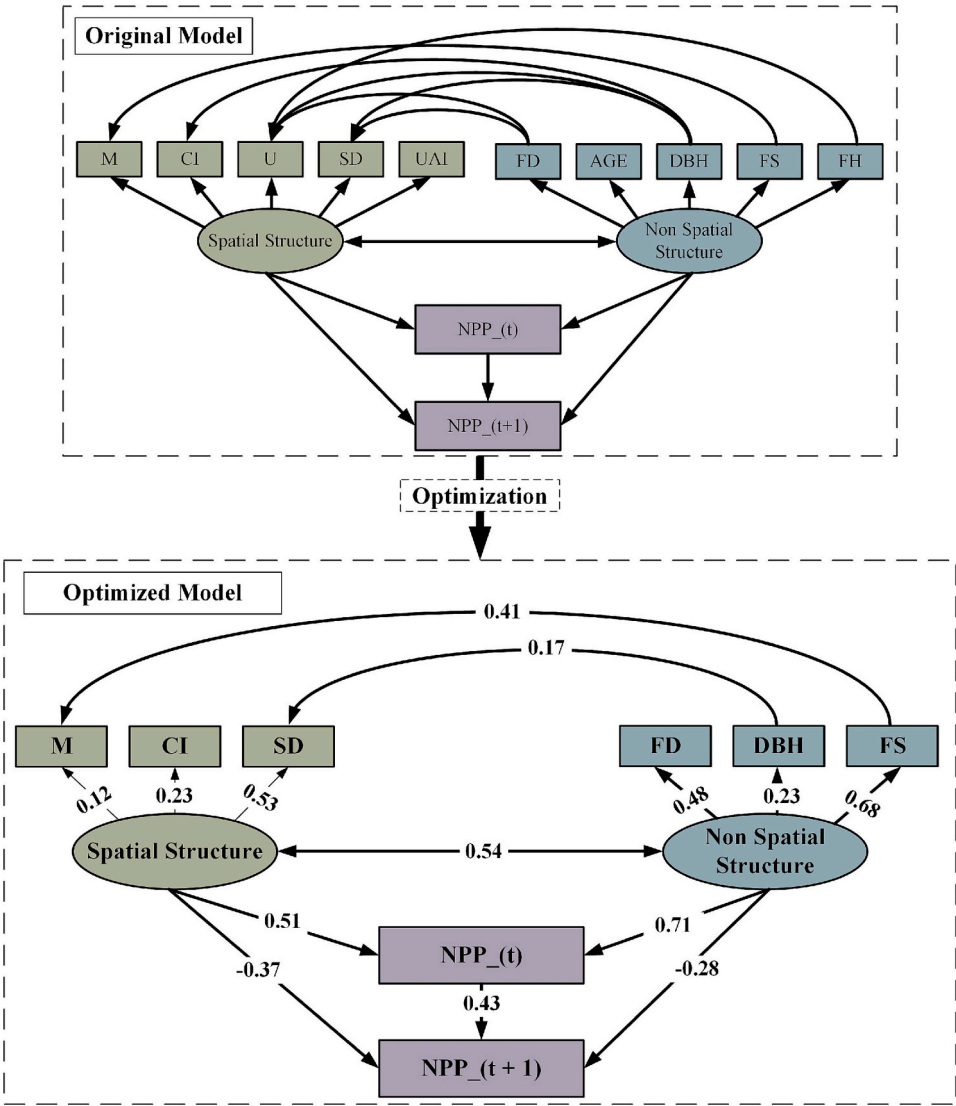


Fig. 4. Original and Optimized Dynamic Structural Equation Models.

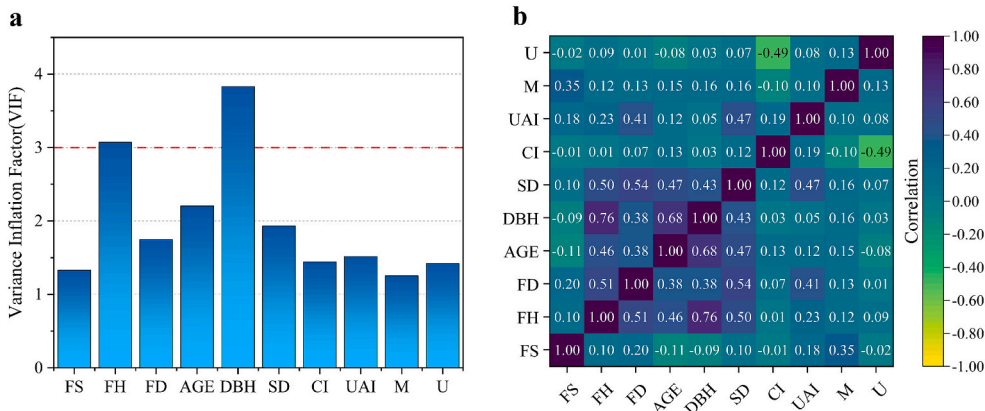


Fig. 5. Collinearity and correlation analysis results among forest structure factors.

directly affect tree growth and physiological processes, which manifest in changes in stand structure factors such as DBH and tree height. These structural modifications are not merely incidental but play a crucial role in determining how forest ecosystems allocate resources and utilize energy. McCarthy and Enquist (2007) argue that optimal partitioning

theory, which suggests that trees allocate resources in ways that maximize growth efficiency, can help explain how stand structure factors, such as tree height and DBH, are shaped by climatic conditions. This theory helps to clarify why variations in stand structure, such as the distribution of tree sizes and canopy characteristics, significantly impact

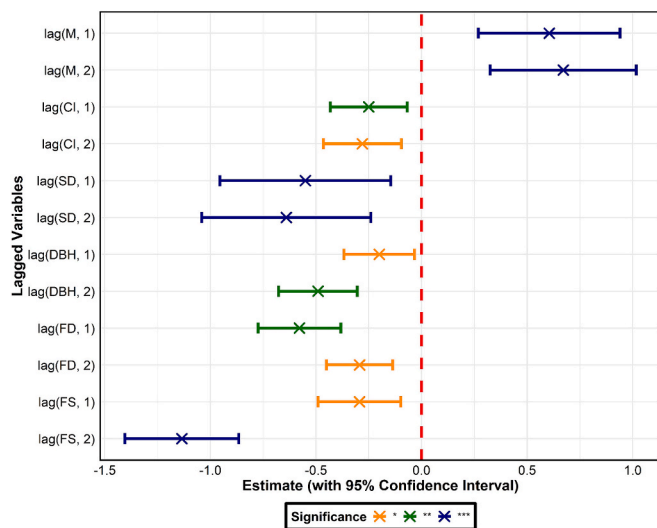


Fig. 6. The significance of the lag effect of stand structure on forest NPP. Lag (x,1) and Lag(x,2) indicate that changes in stand structure factors at time t affect forest NPP at time $t + 5$ and $t + 10$, respectively (with observations in this study recorded every 5 years).

forest productivity. Gao et al. (2014) further support this view by pointing out that climatic factors, particularly temperature and precipitation, are the main drivers of forest biomass and carbon storage, and that significant differences in carbon storage exist among plantations of different tree species. Mensah et al. (2016) add another viewpoint to this discussion by emphasizing that specific functional traits, such as tree size, directly affect biomass allocation between foliage and woody structures. These traits, shaped by both internal (stand structure) and external (climate) factors, ultimately determine the efficiency of resource use in the forest ecosystem. The allocation of biomass, influenced by stand structure, reflects the forest's ability to capture and store carbon, thereby influencing overall forest productivity. Cai et al. (2023) demonstrated that climatic factors, especially mean annual temperature, play a dominant role in shaping tree species richness, whereas stand structure, including tree height variation and maximum tree size, is the main driver of stand biomass in subtropical forests of China. These findings align with our results, which suggest that forest species composition, as a component of stand structure, is a critical factor influencing forest productivity. Similarly, the study by Lv et al. (2024) indicates that forest age and tree size significantly impact the allocation of biomass and carbon storage. As forest age increases, biomass and carbon storage gradually rise, but the allocation patterns (such as the ratio of stem, branches, and leaves) change. This also suggests that forest structure influences forest productivity by affecting resource allocation efficiency, carbon fixation capacity, and ecosystem stability. In summary, these studies highlight the critical role of stand structure, both spatial and non-spatial, in shaping forest NPP. The interplay between climatic conditions, stand structure factors, and resource allocation is a dynamic process, with external factors indirectly influencing NPP through their effects on stand structure. This supports our findings that changes in stand structure, driven by climate and other environmental factors, are key determinants of forest productivity and carbon storage.

Our results indicate that FS is the primary factor influencing forest NPP. Similarly, Mthembeni et al. (2022) found significant variations in NPP among forests with different species compositions. For example, forest with deciduous species like *Acacia* and *Dalbergia obovate* exhibits higher productivity. Both studies highlight the crucial role of species composition in driving NPP, suggesting that variations in forest species significantly impact productivity levels. Seidel et al. (2013) studied the relationship between tree species richness, canopy space use, and forest productivity, and found that tree species diversity helps to enhance

forest productivity. Augusto & Boča (2022) explored how tree functional traits such as forest biomass, and tree species diversity interact with site conditions to drive forest soil carbon storage, which directly supports the impact of tree species composition on forest stability and biomass. Li et al. (2023) examined the role of plant diversity and large-diameter trees in ecosystem functions in tropical evergreen broad-leaved forests, indicating that both large-diameter trees and species diversity significantly affect forest productivity and stability. Tama et al. (2023) demonstrated that tree species diversity promotes forest productivity by enhancing structural complexity, a trend that is consistent across different mycorrhizal types, further emphasizing the key role of tree species composition in forest ecosystems. The studies by Seidel, Augusto, Li, and Tama provide strong evidence that tree species composition is critical for forest stability, as well as for maintaining biomass and enhancing productivity. Our study similarly reveals that both spatial and non-spatial stand structure factors significantly influence forest NPP, with non-spatial factors having a greater impact. This aligns with Ouyang et al. (2019), who also highlighted stand structure as a more influential factor than diversity in regulating forest productivity, noting that stand density plays a more important role than age and maximum DBH, consistent with our findings from Multiple Machine Learning-SHAP.

The results of this study indicate that the influence of stand structure on forest NPP is variable and exhibits a lag effect. Although there is limited research directly addressing the lag effect of stand structure on forest NPP, many studies have explored the impact of external factors on forest NPP. For instance, Xi et al. (2023) found that the effects of terrain, land use type, and climate on forest NPP vary with years. Similarly, Sherry (2008), Kuzyakov (2010), and Li (2023) reported notable lag effects in vegetation growth responses to climate changes. Specifically, Sherry et al. (2008) observed that experimental warming and doubled precipitation in a tallgrass prairie led to delayed increases in above-ground biomass production, highlighting a time lag in vegetation's response to climate changes. This delay underscores the complex, temporal nature of ecosystem responses to climate change. Similarly, Kuzyakov and Gavrichkova (2010) reviewed the time lag between photosynthesis and carbon dioxide efflux from soil, demonstrating that changes in plant growth and carbon cycling do not occur instantaneously when environmental conditions change. Their review suggests that microbial and root activity in the soil introduce significant delays, complicating our understanding of ecosystem feedback to climate change. Li et al. (2023) focused on the time-lag effects in modeling Net Primary Productivity (NPP), showing that incorporating lag effects in light-use efficiency models improves NPP simulation accuracy. Their study emphasizes that climate-induced shifts in vegetation growth are not immediately reflected in ecosystem productivity, underscoring the importance of considering these delays when modeling vegetation dynamics under changing climatic conditions. Similarly, Liu et al. (2024) found a lag effect of surface water and soil water on plant growth in the Yangtze River Basin, illustrating the significant but delayed impact of water availability on vegetation productivity. Additionally, the energy and carbon obtained from photosynthesis take time to be released through respiration to support plant growth (Owen and Mark, 2003). These findings suggest that external factors influence forest NPP with variability and lag. As discussed mentioned above, terrain, soil, and climate indirectly affect forest NPP through their impact on stand structure, aligning with our conclusion that the influence of stand structure on forest NPP is variable and lagged.

In light of these findings, it is essential for forest management measures to account for the temporal nature of ecosystem responses and the variability introduced by stand structure. Forest management measures should prioritize tree species selection, as it has the most significant impact on forest NPP. To maximize forest NPP while maintaining site adaptability to tree species, priority should be given to species exhibiting high productivity, high resource-use efficiency, and strong climate resilience. Optimization of species composition and the establishment of

mixed-species forest ecosystems can significantly enhance both ecological stability and sustainable productivity. Increasing species diversity within stands improves resource utilization efficiency, ecosystem resilience, and overall forest productivity. For example, the study by Cai et al. (2023) highlighted that species diversity, alongside stand structure, plays a critical role in driving forest productivity. A mixed-species forest can better utilize available resources and adapt to environmental changes, supporting higher biomass production with time. Combining species with complementary functional traits (e.g., nitrogen-fixing species with fast-growing species) can further enhance forest growth and resilience to climate change. Optimizing stand density is another crucial measure for maintaining forest productivity. Adjusting stand density through appropriate thinning practices can reduce tree's competition for resources, allowing trees to access sufficient water, light, and nutrients, which are essential for growth and carbon storage. Proper density management not only stabilizes forest productivity but also improves long-term carbon sequestration capacity. Additionally, considering forest age structure in management planning can significantly influence biomass allocation and carbon storage (Lv et al., 2024). A balanced approach that integrates mixed-age stands, including both young and mature trees, can optimize carbon fixation capacity and maintain forest productivity. Finally, enhancing structural complexity, such as increasing tree height variation and canopy layering, can improve light interception and resource allocation efficiency, thereby further boosting forest productivity. Although the implementation of mixed forest configurations can enhance species diversity and promote ecosystem stability, its influence on forest NPP remains relatively secondary when compared to the effects of species composition and stand density. Therefore, incorporating structural complexity should complement, rather than replace, species selection and density regulation as primary management measures.

Compared to traditional modelling approach, the SHAP-DSEM framework significantly improves interpretability by identifying not only the key factors influencing NPP but also their lagged effects. For example, SHAP modelling approach highlights forest species as the most influential factor, while DSEM reveals a significant delayed effect of forest species on NPP. This complementary approach ensures a more comprehensive understanding of stand structure dynamics in forest ecosystems. Despite the valuable insights gained from this study, a limitation lies in the relatively short data time span, consisting of only three time points over a 10-year period with a 5-year interval. While a longer time series could provide a more comprehensive understanding of forest NPP dynamics, collecting detailed individual tree-level data at provincial scale poses considerable labour and financial challenges. Nevertheless, many countries adopt a 10-year planning framework for forest management (Food and Agriculture Organization of the United Nations, 1998). In many countries, including Germany, Finland, Sweden, France, Canada, the United States, China, Japan, and Australia, forest management plans are often formulated and adjusted over multi-year cycles, with a 10-year period being commonly referenced. Therefore, investigating forest stand dynamics over a 10-year time span provides valuable insights for mid-term forest management planning and contributes to a broader understanding of long-term ecological processes. Moreover, our findings highlight the lagged effects of stand structure on NPP, reinforcing the importance of considering temporal dynamics even within shorter time spans. Additionally, this study relied on point data from plot surveys without incorporating external factors such as climate, soil, or terrain. Since climate and soil directly influence key stand attributes like tree height and DBH, which in turn affect forest NPP, their exclusion may introduce uncertainties into the analysis. Future research should integrate climate, soil, and terrain variables to explore their interactions with stand structure and better capture the complex mechanisms driving forest NPP with time. To further validate these findings and explore long-term dynamics, future research should incorporate extended datasets covering longer monitoring periods. Combining long-term monitoring with external environmental variables

would enhance the understanding of how stand structure interacts with broader ecological processes, ultimately improving the quality of forest management plan and the accuracy of ecosystem modeling.

5. Conclusion

To address the research gap in understanding the dynamic and lagged effects of stand structure on forest NPP, this study introduces a new variable selection method using Multiple Machine Learning-SHAP Models and a Dynamic Structural Equation Model (DSEM) to identify the key drivers of forest NPP. This combination of methods allows for a comprehensive analysis of both the immediate and delayed impacts of stand structure on NPP. The main conclusions are as follows:

Impact of Stand Structure: The Multiple Machine Learning-SHAP Models demonstrate that stand structure (both spatial and non-spatial factors) significantly influences forest NPP. Among spatial structure factors, the Stand Density (SD) is the most important, while Forest Species (FS) is the most significant non-spatial factor.

Magnitude of Spatial and Non-Spatial Structural Impacts: The DSEM analysis reveals that non-spatial structural factors exert a more substantial influence on NPP compared to spatial structural factors.

Lagged Effects of Stand Structure: Over time, the influence of stand structure on forest NPP exhibits lagged and dynamic effects, with variations in both the degree and direction of impact.

This study clarifies the impact of stand structure factors on NPP, reveals that the response of forest ecosystems to changes in stand structure has a time delay, deepening our understanding of the driving factors of forest productivity. These findings fill a significant knowledge gap in the field of forest management and provide valuable empirical support for refining ecosystem modeling frameworks. They also provide scientific evidence for optimizing forest management, enhancing carbon sequestration, and addressing climate change.

CRediT authorship contribution statement

Tao Li: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Yi Wu:** Writing – review & editing, Validation, Supervision. **Fang Ren:** Writing – review & editing. **Lei Tian:** Writing – review & editing, Supervision. **Mingyang Li:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Mingyang Li reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper].

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2025.110427>.

Data availability

The authors do not have permission to share data.

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